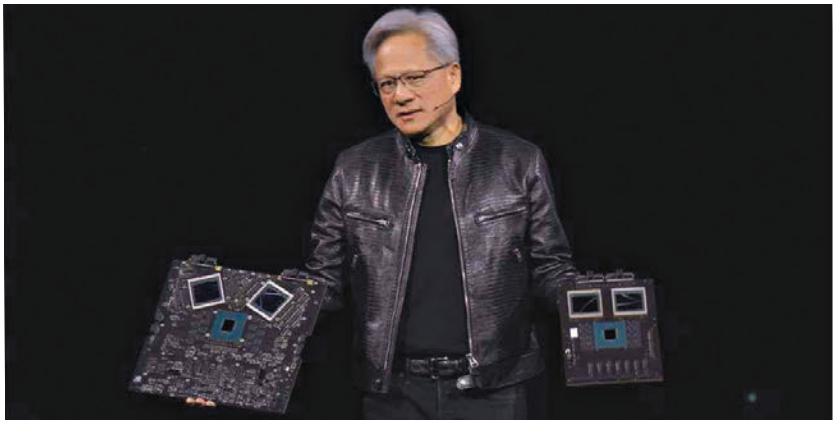

NVIDIA CEO Jensen Huang holding up Blackwell B200 tray

NVIDIA DGX SuperPOD with Blackwell GPUs

B200 GPUs but it should be noted that while all previous claims have been with FP8 precision, NVIDIA's 20 petaflops is on a new FP4 precision. If we are to be fair, then the Blackwell B200's compute performance would be around 10 petaflops with FP8. That's more comparable to what most manufacturers use to indicate compute performance. And with FP8, the Blackwell B200 is only 2.5x times the performance of the H100 does 4 petaflops at FP8. All in all, NVIDIA's B200 isn't that big of a leap as the H100 as the H100 was over its predecessor. And most of it comes down to the limits of the process node.

Does that mean that the FP4 figure is nothing but hot air? Not quite. NVIDIA is using a new second gen Transformer Engine that can help convert AI models to utilise FP4 precision. And the NVIDIA Blackwell GPUs also support a new FP6 format as a go-between that offers a bit more precision than FP4 for models that can make do with slightly less precision than FP8.

Overall, NVIDIA is claiming to get a 4x increase in AI training performance and 30x increase in inference performance compared to the NVIDIA H100. Nvidia offers various Blackwell configurations, including the GB200 superchip with two B200 GPUs and a Grace CPU, offering up to 20 petaflops of FP4 AI inference per GPU. The HGX B200 and B100 configurations cater to different power and performance requirements, with the latter designed for compatibility with existing H100 infrastructure.

NVIDIA GB200 NVL72 Spire

NVLink 7.2T

With new GPUs come faster interlinks. You have to put in faster interconnects otherwise all the compute improvements are of no use as they'd get bottlenecked. And interconnects are very expensive in terms of power and time, as resources. With Blackwell, NVIDIA is bringing out the fifth generation of NVLink and along with it, a new NVLink 7.2T switch.

The new NVIDIA NVLink 7.2T chip supports 7.2 TB/s of full-duplex bandwidth cumulatively. Each fifth generation NVLink interface has a max bandwidth of 1.8 TB/s and with four such NVLinks on one chip, the net bandwidth becomes 7.2 TB/s. The NVLink 7.2T chip is also built using TSMC's 4NP process node and it supports 3.6 teraflops of Sharp v4 in-network compute on chip. SHARP (Scalable Hierarchical Aggregation and Reduction Protocol) comes from Mellanox which was acquired by NVIDIA. It is a technology that improves performance of large scalable HPC machines by offloading collective operations from CPUs and GPUs to networking chips.

NVIDIA B200 NVL72

All of NVIDIA's announcements clubbed together brings you the NVIDIA B200 NVL72, a massive GPU which is made of 18 1U servers that have two GB200 superchips, each. Each superchip is built using a single Grace CPU linked to two GB200 GPUs. This is unlike H100 which had a 1x CPU-1x GPU pairing. Each server will have two Grace GPUs and four NVIDIA Blackwell GPUs. That's about 80 petaflops of FP4 performance. Each tray will also have two NVLink switches giving rise to a cumulative bandwidth of 14.4 TB/s.

Overall, the NVIDIA GB200 NVL72 has 72 Blackwell GPUs, hence the 72 in the name. Along with that it gets 36 Grace CPUs, and 130 TB/s of cumulative bandwidth thanks to 18 NVLink switches. AT GTC 2024, NVIDIA also announced the SuperPOD which allows 576 GB200 Blackwell GPUs per domain. d